In each of Problems 1 through 12, determine the general solution of the given differential equation that is valid in any interval not including the singular point.

$x=0$

5.4. 9. $x^2y'' - 5xy' + 9y = 0$

$y = x^r \quad y' = r x^{r-1} \quad y'' = r(r-1)x^{r-2}$

$$x^r \underbrace{\left[r(r-1) - 5r + 9\right]}_{F(r)} = 0$$

$$F(r) = r^2 - r - 5r + 9 = r^2 - 6r + 9 = (r-3)^2 = 0$$

$r_{1,2} = 3$ equal roots

$y_1 = c_1 x^3 \qquad y_2 = c_2 x^3 \ln|x|$

gen. sol.: $y = c_1 x^3 + c_2 x^3 \ln|x|$ on $(-\infty, 0)$ OR $(0, \infty)$

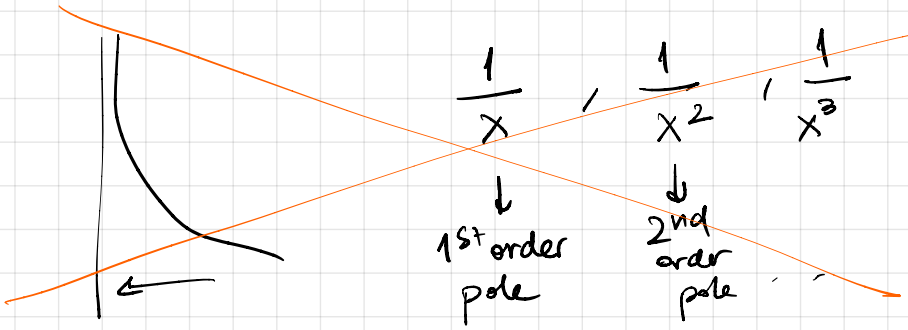
$y = |x|^3 = (-x)^3$ on $(-\infty, 0)$

In each of Problems 17 through 34, find all singular points of the given equation and determine whether each one is regular or irregular.

19. $\underbrace{x^2(1-x)}_{A(x)}y'' + \underbrace{(x-2)}_{B(x)}y' \underbrace{- 3x}_{C(x)}y = 0$

$1y'' + \frac{B(x)}{A(x)}y' + \frac{C(x)}{A(x)}y = 0$

x_0 singular if $A(x)=0 \longrightarrow x_0 = 0, 1$



(1) $\lim_{x \to x_0} \frac{B(x)}{A(x)} \cdot (x - x_0) = ? \quad < \infty$

(2) $\lim_{x \to x_0} \frac{C(x)}{A(x)} \cdot (x - x_0)^2 = ? \quad < \infty$

} regular singularity

$x = 0$

$\lim_{x \to 0} \frac{x-2}{x^2(1-x)} \cdot x = \infty \rightarrow$ irregular singularity (1) ✗

$x = 0$

$x = 1$

$\lim_{x \to 1} \frac{x-2}{x^2(1-x)} (x-1) = \lim_{x \to 1} \frac{2-x}{x^2} = 1$ ✓

$\lim_{x \to 1} \frac{-3x}{x^2(1-x)} (x-1)^2 = \lim_{x \to 1} \frac{3(x-1)}{x} = 0$ ✓

$x = 1$ regular singularity

In each of Problems 41 and 42, show that the point $x = 0$ is a regular singular point. In each problem try to find solutions of the form $\sum_{n=0}^{\infty} a_n x^n$. Show that (except for constant multiples) [redacted] there are no nonzero solutions of this form in Problem 42. Thus in neither case can the general solution be found in this manner. This is typical of equations with singular points.

42. $2x^2y'' + 3xy' - (1+x)y = 0$

$M(x)$ $N(x)$ $O(x)$

$x = 0$ singular since $M(x) = 2x^2 = 0$ if $x = 0$.

(1) $\frac{N(x)}{M(x)}$ ~~better than 1st order pole singularity?~~

$\lim_{x \to 0} \frac{3x}{2x^2} \cdot x = \frac{3}{2}$ ✓

(2) $\frac{O(x)}{M(x)}$ ~~better than 2nd order pole singularity?~~

$$\lim_{x\to 0} \frac{-(1+x)}{2x^2}\, x^2 = -\frac{1}{2} \quad \checkmark$$

$$y = \sum_{n=0}^{\infty} a_n x^n \qquad y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \qquad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$2x^2 \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + 3x \sum_{n=1}^{\infty} n a_n x^{n-1} - (1+x) \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} 2n(n-1) a_n x^n + \sum_{n=1}^{\infty} 3n a_n x^n - \sum_{n=0}^{\infty} a_n x^n - \underbrace{\sum_{n=0}^{\infty}}_{k=1} a_n x^{\overbrace{n+1}^{k}} = 0 \qquad \downarrow k-1$$

$$\sum_{n=2}^{\infty} 2n(n-1) a_n x^n + \sum_{n=1}^{\infty} 3n a_n x^n - \sum_{n=0}^{\infty} a_n x^n - \sum_{n=1}^{\infty} a_{n-1} x^n = 0$$

$$3a_1 x - a_0 - a_1 x - a_0 x + \sum_{n=2}^{\infty} \left[2n(n-1) a_n + 3n a_n - a_n - a_{n-1} \right] x^n = 0$$

$-a_0 = 0 \rightarrow a_0 = 0$ (const. terms)

$3a_1 - a_1 - a_0 = 0 \rightarrow a_1 = 0$ (coef. of x)

$\forall n \geq 2$: $2n(n-1) a_n + 3n a_n - a_n - a_{n-1} = 0$

$$\left(2n(n-1) + 3n - 1\right) a_n = a_{n-1}$$

$$a_n = \frac{1}{2n(n-1) + 3n - 1}\, a_{n-1}$$

$a_0 = 0$
$a_1 = 0$
$\vdots$
$a_k = 0$
$\vdots$

$$y = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 0\, x^n = 0$$

5.5.4) $x=0$

$$xy'' + y' - y = 0 \quad \cdot \frac{1}{x}$$

$$y'' + \frac{1}{x}y' - \frac{1}{x}y = 0$$

$\lim_{x\to 0} x\cdot\frac{1}{x} = 1$ $\quad$ $\lim_{x\to 0} x^2\cdot\frac{-1}{x} = 0$ $\quad\Rightarrow$ $x=0$ regular sing. pt.

$1 = \underset{p_0=1}{1} + \underset{p_1=0}{0}x + 0x^2 + \dots$ $\qquad -x = 0 - x + 0x^2 \dots$

$$y = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{r+n} = a_0 x^r + a_1 x^{r+1} + \dots$$

$$y' = \sum_{n=0}^{\infty} (r+n) a_n x^{r+n-1}$$

$$y'' = \sum_{n=0}^{\infty} (r+n)(r+n-1) a_n x^{r+n-2}$$

$$x\sum_{n=0}^{\infty} (r+n)(r+n-1) a_n x^{r+n-2} + \sum_{n=0}^{\infty} (r+n) a_n x^{r+n-1} - \sum_{\substack{n=0 \\ k=1}}^{\infty} a_n x^{r+n} = 0 \qquad k-1=n,\ a_n \to a_{k-1}$$

$$\sum_{n=0}^{\infty} (r+n)(r+n-1) a_n x^{r+n-1} + \sum_{n=0}^{\infty} (r+n) a_n x^{r+n-1} - \sum_{n=1}^{\infty} a_{n-1} x^{r+n-1} = 0$$

$$r(r-1)a_0 x^{r-1} + r a_0 x^{r-1} + \sum_{n=1}^{\infty} \left[(r+n)(r+n-1)a_n + (r+n)a_n - a_{n-1}\right] x^{r+n-1} = 0$$

↳ ① x^{r-1} term: $\quad r(r-1)a_0 + r a_0 = 0 \Rightarrow \boxed{r^2 a_0 = 0}$ (indicial eqn.) $\to r_{1,2} = 0$

② x^{r+n-1} terms: $\quad (r+n)(r+n-1)a_n + (r+n)a_n - a_{n-1} = 0$

$$a_n (r+n)^2 = a_{n-1} \Rightarrow \boxed{a_n = \frac{a_{n-1}}{(r+n)^2}}$$

recurrence relation

$r=0$: $a_n = \dfrac{a_{n-1}}{(r+n)^2} \rightarrow a_n = \dfrac{a_{n-1}}{n^2}$

recurrence relation

a_0, $a_1 = \dfrac{a_0}{1}$, $a_2 = \dfrac{a_1}{2^2} = \dfrac{a_0}{2^2}$, $a_3 = \dfrac{a_2}{3^2} = \dfrac{a_0}{2^2 3^2}$

$a_4 = \dfrac{a_0}{2^2 3^2 4^2}$ ....

$\left(2 \cdot 3 \cdot 4\right)^2$

$a_k = \dfrac{a_0}{(k!)^2}$

$y = \sum_{n=0}^{\infty} a_n x^{r+n} = a_0 + \sum_{n=1}^{\infty} \dfrac{a_0}{(n!)^2} x^n \rightarrow \lim_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \dfrac{(n!)^2}{[(n+1)!]^2}$

$= \lim_{n\to\infty} \dfrac{1}{(n+1)^2} = 0$

$\Rightarrow$ converges for all x.

5.5.13 $x y'' + (1-x) y' + \lambda y = 0$ $\quad \frac{1}{x}$

a) $x=0$ reg. sing. pt. ✓

b) $y = \sum_{n=0}^{\infty} a_n x^{n+r}$

$\vdots$

(1) x^{r-1} term: $r^2 a_0 = 0 \rightarrow$ indicial $\boxed{r^2 = 0} \rightarrow r_{1,2} = 0$

x^{r+n-1} terms: $(r+n)^2 a_n - (r+n-1-\lambda) a_{n-1} = 0$

$a_n = \dfrac{(r+n-1-\lambda)}{(r+n)^2} a_{n-1}$ recurrence

$r=0$ $\quad a_n = \dfrac{(r+n-1-\lambda)}{(r+n)^2}\, a_{n-1} \xrightarrow{r=0} a_n = \dfrac{n-1-\lambda}{n^2}\, a_{n-1}$

$a_0,\ a_1 = \dfrac{-\lambda}{1}\, a_0$ (=0 for $\lambda=0$), $\quad a_2 = \dfrac{1-\lambda}{2^2}\, a_1 = \dfrac{(1-\lambda)(-\lambda)}{2^2}\, a_0$ (=0 if $\lambda=1$)

$$a_3 = \frac{(2-\lambda)(1-\lambda)(-\lambda)}{2^2 \cdot 3^2}\, a_0 \quad (=0 \text{ if } \lambda=2)$$

$$a_n = \frac{\prod_{k=0}^{n-1} (k-\lambda)}{(n!)^2}\, a_0 \qquad \left((n-1)-\lambda\right)\cdots(2-\lambda)(1-\lambda)(0-\lambda)$$

If $\lambda = m \in \mathbb{Z}^+$, $a_{m+1} = 0$.

$\Downarrow$

y = a polynomial of degree m

$$y = a_0 + \frac{-\lambda}{1} a_0 x + \frac{(1-\lambda)(-\lambda)}{2^2} a_0 x^2 + \cdots$$

5.6.16 $\quad xy'' + y = 0$ $\quad$ a) $x=0$ reg. sing. pt.

$$y = \sum_{n=0}^{\infty} a_n x^{r+n}$$

$$y' = \sum_{n=0}^{\infty} (r+n) a_n x^{r+n-1}$$

$$y'' = \sum_{n=0}^{\infty} (r+n)(r+n-1) a_n x^{r+n-2}$$

$$\times \sum_{n=0}^{\infty} (r+n)(r+n-1) a_n x^{r+n-2} + \sum_{n=0}^{\infty} a_n x^{r+n} = 0$$

$$\sum_{n=0}^{\infty}(r+n)(r+n-1)\,a_n x^{r+n-1} + \sum_{n=1}^{\infty} a_{n-1} x^{r+n-1} = 0$$

$F(r)=0$, $F(1)=0$

$a_n \cdot F(0+n) = -a_{n-1}$, $a_k F(r_2+k) = -\Sigma[\,]a_{k-1}$, $M=1$

$a_1 F(1) = -a_0$, $F(r_1)=0$ → consistency condition NOT satisfied

$$\underbrace{r(r-1)\,a_0\,x^{r-1}}_{\text{indicial eqn.}} + \sum_{n=1}^{\infty}\underbrace{\left[\overbrace{(r+n)(r+n-1)}^{F(r+n)}\,a_n + a_{n-1}\right]}_{\text{recurrence relation}} x^{r+n-1} = 0, \quad n \geq 1$$

$r(r-1)=0$

$r_1 = 0$, $r_2 = 1$ → exponents at $x=0$

$r_2 = 1$: bigger root

$$a_n = \frac{-1}{(r+n)(r+n-1)}\,a_{n-1}$$

$$\underline{r=1}: \quad a_n = \frac{-1}{(1+n)(1+n-1)}\,a_{n-1} = -\frac{a_{n-1}}{n(n+1)}$$

$$a_0, \quad a_1 = -\frac{a_0}{1\cdot 2}, \quad a_2 = -\frac{a_1}{2\cdot 3} = \frac{a_0}{1\cdot 2\cdot 2\cdot 3}$$

$$a_3 = -\frac{a_2}{3\cdot 4} = -\frac{a_0}{1\cdot 2\cdot 2\cdot 3\cdot 3\cdot 4}$$

$$a_k = (-1)^k \frac{a_0}{k!\,(k+1)!}$$

$$y_1 = x^{\overset{\uparrow}{r}} \sum_{n=0}^{\infty} a_n x^n = x \sum_{n=0}^{\infty} (-1)^n \frac{a_0}{n!\,(n+1)!}\,x^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{a_0}{n!\,(n+1)!}\,x^{n+1}$$

cy_1 is also a solution

$x>0$

Consistency check ⇒ NOT satisfied ⇒ ln term:

$$y_2 = a y_1 \ln|x| + |x|^0 \left[1 + \sum_{n=1}^{\infty} c_n x^n\right]$$

$$= a y_1 \ln x + \left[1 + \sum_{n=1}^{\infty} c_n x^n\right] \quad \left[= 1 + c_1 x + c_2 x^2 \ldots\right]$$

$$y_2' = a y_1' \ln x + a y_1 \frac{1}{x} + \sum_{n=1}^{\infty} n c_n x^{n-1}$$

$$y_2'' = a y_1'' \ln x + a y_1' \frac{1}{x} + a y_1' \frac{1}{x} - a y_1 \frac{1}{x^2} + \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$x\left[a y_1'' \ln x + a y_1' \frac{1}{x} + a y_1' \frac{1}{x} - a y_1 \frac{1}{x^2} + \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}\right] + a y_1 \ln x + \left[1 + \sum_{n=1}^{\infty} c_n x^n\right] = 0$$

$$\ln x \left[a y_1'' x + a y_1\right] - a y_1 \frac{1}{x} + 2 a y_1' + \sum_{n=2}^{\infty} n(n-1) c_n x^{n-1} + 1 + \sum_{n=1}^{\infty} c_n x^n = 0$$

$xy'' + y = 0$ & y_1 is a soln. ⇒ $a y_1'' x + a y_1 = 0$

$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-1}$: $k = n-1$, $n = k+1$

$$-\frac{1}{x} a \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!(n+1)!} x^{n+1} + 2a \sum_{n=0}^{\infty} (-1)^n \frac{n+1}{n!(n+1)!} x^n = -1 - \sum_{k=1}^{\infty} (k+1) k c_{k+1} x^k - \sum_{k=1}^{\infty} c_k x^k$$

$$\sum_{k=0}^{\infty} \left[\frac{-a(-1)^k}{k!(k+1)!} + \frac{2a(-1)^k (k+1)}{k!(k+1)!}\right] x^k = -1 + \sum_{k=1}^{\infty} \left[-(k+1) k c_{k+1} - c_k\right] x^k$$

$$= \frac{a(-1)^k (-1 + 2k + 2)}{k!(k+1)!}$$

$$= \frac{a(-1)^k (2k+1)}{k!(k+1)!}$$

$k=0$: $-a + 2a = -1 \rightarrow a = -1$ (not necessary; just put $k=0$)

$k \geq 1$: $\dfrac{a(-1)^k(2k+1)}{k!\,(k+1)!} = -(k+1)k\,c_{k+1} - c_k$

$$c_{k+1} = \left[-\frac{a(-1)^k(2k+1)}{k!\,(k+1)!} - c_k \right] \frac{1}{k(k+1)}$$

There is no constraint for c_1 $\Rightarrow$ choose a $c_1 \in \mathbb{R}$. Calculate others.

Choosing c_1 in a different way is equivalent to adding/substracting $c_1 y_1$ to the second solution. Check it!